

The circuit diagram illustrates the connection of a CH224K IC, which acts as a USB-to-I2C bridge. The IC's pins are connected as follows:

- VBUS**: Connected to the VBUS pin of the USB connector J1 through a 3 A fuse F1.
- GND**: Connected to the GND pin of the USB connector J1.
- CC1** and **CC2**: Connected to the CC1 and CC2 pins of the USB connector J1.
- D+** and **D-**: Connected to the D+ and D- pins of the USB connector J1.
- P0**: Connected to the PD_PG pin of the I2C bus.
- CFG1**, **CFG2**, and **CFG3**: Connected to the CFG1, CFG2, and CFG3 pins of the I2C bus.

A note at the bottom states: "CH224K requests 5 / 9 / 12 / 15 / 20 V through CFG1...3. The MCU re-reads ADC_VBUS 500 ms after plug-in; no rise means a legacy 5 V port."

[illegible]

NPN Darlington emitter follower – the op-amp drives the base, the emitter follows. Shunt in the output return, second LM358 half lifts it into ADC range.

LP2950-5.0

The diagram shows the LP2950-5.0 voltage regulator circuit. The regulator is represented by a yellow box with pins labeled VI (pin 3), V0 (pin 1), and GND (pin 2). The input pin VI is connected to a green line labeled 'VBUS' and a 10uF capacitor (C6) to ground. The output pin V0 is connected to a green line that leads to a 5V output terminal and a 10uF capacitor (C7) to ground. The GND pin is connected to a ground symbol. A red arrow indicates the output voltage is 5V.

Diagram of the J5 connector pinout:

- Pin 1: SWIO
- Pin 2: UART_TX
- Pin 3: UART_RX
- Pin 4: Ground

github.com/mohammadrezasafaeian (c) 2026 Mohammadreza Safaeian. All rights reserved. Designed and drawn by Mohammadreza Safaeian - m.re.safaeian@gmail.com Single-sheet design - v0.3 Mohammadreza Safaeian - personal project		
Sheet: / File: usbc_pd_psu.kicad_sch		
Title: USB-C PD Bench PSU - 4-Switch Buck-Boost + PNP Linear		
Size: A3 KiCad E.D.A. 10.0.3	Date: 2026-03-24	Rev: v0.3 Id: 1/1